

MA 301

Test 1

Fall - 2011

Problem	points	out of
1		10
2		10
3		10
4		10
5		10
Score		50

Name: _____

HR: _____

Signature: _____

READ BEFORE YOU START

- * Show all work.
- * Change your calculator mode to RADIANS.
- * Sloppy solutions may be penalized.
- * Use calculators ONLY for computations.
- * You must write up every problem in a clear and organized way.
- * If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1: Given $\mathbf{F} = [2, 2y + z, y + 1]$

(a) Show that $\mathbf{F}$ is conservative in two different ways

method 1

method 2

(b) Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ when C is a straight line from $(-1, 4, 2)$ to $(0, 0, -6)$

Problem 2: Let $\mathbf{F} = [x + y, 2xy]$ and C be the boundary of the region bounded by $y = x^2$ and $y = x + 2$ in the x, y -plane.

(a) Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$ using green's theorem.

(b) Setup the integral you would use to compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$ directly (*don't leave answer as a dot product*)

Problem 3: Let $\mathbf{F} = [x, y, z]$ and S be the sphere $x^2 + y^2 + z^2 = 25$. Compute $\iint_S \mathbf{F} \cdot \mathbf{n} \, dA$

Problem 4: Compute the flux of water (water density = $1 \text{ ton}/m^3$) through the parabolic cylinder $S : y = x^2$ where $0 \leq x \leq 2$ and $0 \leq z \leq 1$ if the velocity flow of water is given by $\mathbf{v} = [3z^2, 6, 6xz]$ (units: m/s).

Problem 5: Answer the following conceptual questions

(a) Suppose $\mathbf{F} = [1, 1, 1]$. Is it true that $\iint_S \mathbf{F} \cdot \mathbf{n} \, dA = 0$ for any closed surface S which is a boundary of a solid? Justify your answer.

(b) Describe the difference between Green's Theorem and Stokes' Theorem.

(c) Briefly describe the meaning of flux.

(d) Let $\mathbf{F}$ vector field such that $\text{curl } \mathbf{F} = \mathbf{0}$. Match each expression in the left column with the most appropriate term in the right column.

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

conservative

$$\mathbf{F} \cdot d\mathbf{r}$$

zero

$$\oint_C \mathbf{F} \cdot d\mathbf{r}$$

path independent

$$\mathbf{F}$$

exact

(e) Suppose we know to represent a hill as a surface given by the function $f(x, y)$. If we place a ball anywhere on the hill, what direction would you expect it to roll when we let it go?